

Creating Solid Geometry Using BRL-CAD

Anything that happens in real time is easier to conceptualise. This can be well illustrated by 3D games where users find simulation adventurous. It is the thrill of real-time effects that makes these games a hit amongst the masses.

3D simulation can also be a boon for designers who may find designing PCBs pretty boring with a black-and-white screen and 2D images. So simulation designing tools are taking the market by storm. BRL-CAD is one such open source freeware.

3D solid modelling made possible by BRL-CAD

BRL-CAD is like AutoCAD in many aspects. It supports 2D and 3D imaging and runs on Windows, Mac OS and Linux platforms. 3D solid modelling using BRL-CAD software can accelerate design, thus saving time and boosting productivity. 3D view helps designers to conceptualise real-time objects.

BRL-CAD is a constructive solid-geometry and solid-modelling CAD tool that can be used

in a variety of engineering and graphics applications. It includes geometry editor, ray tracing support for graphics rendering and geometric analysis, computer network distributed frame buffer support, scripting, image processing and signal processing tools. It is mainly used for ballistic and electromagnetic analyses.

Features

Geometry editor. BRL-CAD includes features of solid shapes from arbitrary convex polyhedral to ellipsoids, cones, cylinders. Apart from designing solid shapes, it also lets users animate designs without requiring a separate software for visualising 3D solid figures.

Designers can add different curve shapes and deformities in a 3D structure. Apart from assembly modelling, BRL-CAD offers advanced surface features such as curvature analysis, draft analysis, thickness analysis and symmetry check analysis. It also includes tools to calculate area and volume of the shapes. Thus, users can be assured that the results are precise.

After making various 3D parts, combining them to make one perfect product is possible through 3D.

Ray tracing support for graphics rendering and geometric analysis. BRL-CAD uses ray tracing technology to produce a very high degree of virtual realism. This technique can be used in designing where images can be rendered slowly ahead of time, but it cannot

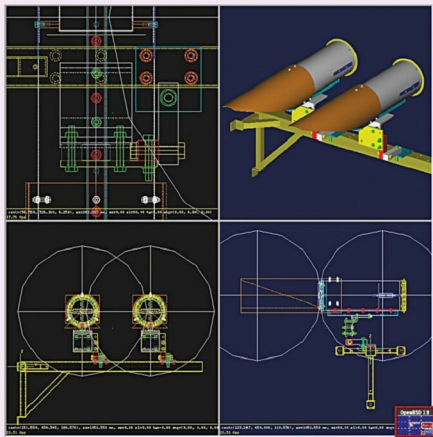


Fig. 1: BRL-CAD work page (Image courtesy: www.upubuntu.com)